



CHAPTER 6

METHODS OF TRAINING

PHOTO: MATT PINCUS

The following section highlights some of my favorite sessions for addressing the needs of the various categories of training introduced earlier in the book. These are organized in general and specific sections for each group. As I noted above, the general sessions will trend toward energy production and larger muscle groups, the specific ones will aim more toward what we do in the performance environment. Once you have a good grasp of what kind of work you should be doing in each style of session, you can create nearly an infinite variety of sessions within each category.

STRENGTH TRAINING SESSIONS

GENERAL STRENGTH SESSIONS

HIGH LOAD STRENGTH TRAINING

Being generally stronger can help your overall conditioning and improve your power in addition to the benefits listed above. Many people fear strength training because they associate the use of weight with bodybuilders, but strength training and hypertrophy training are different activities. In high-load strength training, you need to work at a load that is 80% or more of your maximum ability to do one repetition. To some degree, the prescribed number of reps will help you stay in this zone, but it's wise to test it once in a while to be sure.

This training can be done two times per week for most climbers and deliver progressively better strength over the entire phase. In a high-load session, we plan to cover the major movement patterns plus a hard midsection exercise or two. Remember that fatigue is not the goal here, and therefore you should rest as much as 5 minutes between sets of a given movement. I like to set up the training with forced recoveries in the form of stretching between exercises and/or alternating between two exercises in each set.

The best starting point for these sessions is to do 3 sets of 3 reps for each exercise. You can adjust this slightly as your training develops, adding some more sets for greater capacity, decreasing reps as you push higher loads, etc. At first, this session might go by quickly, but as your loads increase, it will eventually take quite some time. The simple template below, if done with very high loads, can take an athlete much more than an hour. Since we are climbers who lift, not lifters who climb, most of us will finish in 30-45 minutes.

A basic template for this session is as follows:

GROUP A (3 SETS):

Upper Body Pull (Pull-Ups, Dumbbell Rows, Inverted Rows)
Squat (Front Squat, Step-Up)
Core Flexion (Ab Wheel, Hanging Leg Raise)
Mobility (60-90 seconds shoulders or hips)

GROUP B (3 SETS):

Upper Body Press (Overhead Press, Push-Ups, Bench Press, Dips)
Hip Hinge (Deadlift, Trap Bar Deadlift, Lever)
Core Stability (Plank/Pull, Hollow Holds)
Mobility (60-90 seconds shoulders or hips)

Remember that this training can be done on the same day as a finger strength session.

INTEGRATED STRENGTH TRAINING

In my mind, combining finger strength training, mobility work, and resistance training is the best of all worlds. If you're like me, you want to spend as little time as possible gaining strength, and as much time as possible using it. By using combination sessions, you get a whole host of benefits:

- Fewer total training sessions dedicated to strength means more time climbing.
- By adding hangboarding to your resistance training, you are creating longer and more effective rest periods, so you don't end up under-resting between heavy sets. This equals greater strength gains and a lower chance of injury.
- By sticking a resistance set and some stretching between each of your hangs, you reduce injury potential in the fingers and give yourself time to recover between hard sets.
- By integrating your mobility within the session, you'll actually do it.
- Finally, and most important, you'll get stronger fingers with this program than by training fingers alone. I'll explain below.

We get strong because of a couple of factors. Strength comes primarily from improvement in neural factors, as noted above, and increases in strength due to hormonal changes that take place with hard training. Unfortunately, simply training the fingers and forearms in a normal hangboard session is not enough of an overload to your system to cause a significant hormonal change. When we add even a few squats or presses or other heavy strength movements into the finger strength workout, we get not only the

neural stimulus local to the forearm, but we get a global stimulus in the form of a big release of anabolic hormones.

I suspect you could do a heavy strength session, followed by some finger work and get good results. But there is an even better way for the two exercises to compliment each other. I call this *integrated strength*.

The standard Integrated Strength workout format is as follows:

3 SETS OF EACH GROUP, DONE CIRCUIT-STYLE:

A1: Hangboard position 1

A2: Deadlift or Front Squat, 2-5 reps

A3: Hip Mobility, 60 seconds

B1: Hangboard position 2

B2: Push-Up Variation or Overhead Press, 4-6 reps

B3: Shoulder Mobility, 60 seconds

C1: Hangboard position 3

C2: Hanging Straight Leg Raise, Ab Wheel, or Levers, 4-6 reps

C3: Hip Mobility, 60 seconds

(Each group takes about 15 minutes to execute.)

We can use the same integrated strength format with any style of hangboard protocol, but simple sets of 7-10 second hangs seem to work the best. Whatever you choose, You should stick to one protocol, and use the same resistance exercises for an entire 4 week strength phase. A favorite pair of workouts using other protocols is as follows:

Integrated Strength with Hangboard Ladders

3 SETS OF EACH GROUP, DONE CIRCUIT-STYLE:

A1: Half Crimp x3-6-9 (one hang every 20 seconds, 60 seconds total)

A2: Deadlift x3

A3: Fire Log Stretch, 60 seconds

B1: Open Hand x3-6-9 (one hang every 20 seconds, 60 seconds total)

B2: 1-arm, 1-leg Push-Up x3+3

B3: Kettlebell Arm Bar, 30 seconds each side

C1: Full Crimp x3-6-9 (one hang every 20 seconds, 60 seconds total)

C2: Hanging Straight Leg Raise x6

C3: Splits, 60 seconds

Integrated Strength with Single Arm Hangs

3 SETS OF EACH GROUP, DONE CIRCUIT-STYLE:

A1: Hang, Straight Arm x10+10 seconds

A2: Front Squat x5

A3: Frog Stretch, 60 seconds

B1: Hang, Bent Arm x10+10 seconds

B2: 1-Arm Overhead Press x5+5

B3: Doorway Shoulder Opener, 60 seconds

C1: Hang, Lock Off x10+10 seconds

C2: Levers x3

C3: Splits, 60 seconds

The Integrated Strength template is the primary one I recommend you use during your Strength days on a standard nonlinear cycle. It is also appropriate as your strength selection in the General Strength phase of a block plan.



PHOTO: JULIE ELLISON

SPECIFIC STRENGTH SESSIONS

FULL COMBINATION STRENGTH

In the Full Combination workout, we train five different facets of strength in five different movement patterns. The idea here is to break the habit of doing a single tempo in resistance training - which most of us adopt unconsciously. Many of us train the same pace in squats and lunges and presses and pull-ups...and our muscles learn to work according to those rules. When we hit the rock, though, things change. We have to stand in a crouched position for 30 seconds. We have to lunge for a jug. We have to lock off to place our next finger lock. By preparing for this in the gym, we increase our chance of success substantially.

These sessions are built to help you bridge your strength from heavy “normal” exercises, to doing the exercises in ways that are more like you’d encounter in the real world.

MOVEMENT PATTERNS

Upper Body Press

Upper Body Pull

Hip Hinge

Squat

Core

EXERCISE VARIATIONS

- ◇ Bilateral - Exercises are done with both arms or both legs at once, using the same pattern. Examples include pull-ups, squats, deadlifts, and bench press. You can use a barbell or a pair of weights for most variations. This is the most common form of training you see in gyms today.

- ◇ Unilateral - Exercises are done with one limb at a time or using different patterns. Examples include step-ups, single arm presses, dumbbell rows, lunges, and single-leg raises.
- ◇ Isometric - Holding a static position, normally not at the beginning of the range of motion. Exercises include almost anything you could do concentrically, but held against an immovable object: Iso hold pull-ups, presses, squats, planks, etc.
- ◇ Explosive - A power version of a strength exercise. Jumps, explosive push-ups, doubles on the campus board, and medicine ball work all fit the bill. This is the type of strength that dominates in power training phases.
- ◇ Unstable - A destabilized version of an exercise. You can use an uneven surface, unstable surface, or a variable resistance tool such as a sandbag. Exercises can include core work on a stability ball or medicine ball, elastic band work, offset deadlifts or squats, or Airex Pad lunges. These are usually done at lighter loads than the stable exercises above. The value of unstable exercise is obvious to athletes, the error is doing too much. Remember: too little stability = too little force. If everything is destabilized, you will get weak.

The sessions are built on doing one of each exercise variation and one of each movement pattern. This is where your programming can be a bit creative. The important things to keep in mind are as follows:

- This is still strength training. Keep the intensity high, and rest enough between sets to be fresh for the next set.
- Do 2-3 sets per exercise, and 4-6 reps per set. Isometrics should be held for 5-10 seconds.

- Combine this session with a hangboard or climbing session. A Full Combination workout might take 30-40 minutes, which you can then follow with some finger strength. You can also integrate a hangboard session if your facility is set up for it.
- You can do the exercises as one big circuit of five, as a couple of groups, or as single sets combined with mobility or finger strength. As long as you keep pushing strength, it doesn't really matter.

A simple large circuit session might look like this:

3 ROUNDS OF:

Front Squat x5 (squat / bilateral)

Dumbbell Row x5+5 (upper body pull / unilateral)

Double Kettlebell Swing x5 (hip hinge / explosive)

Bench Press Iso Hold x8 seconds (upper body press / isometric)

Sandbag Get-Up x1+1 (core / unstable)

Full Combo Sessions can also be broken down into a bi-set or tri-set format. These are more effective in crowded gyms or if you have limited equipment. An example set-up might look like this:

GROUP 1, 3 ROUNDS:

Step-Up x5+5 (squat / unilateral)

Iso Hold Pull-Up x5 seconds (upper body pull / isometric)

GROUP 2, 3 SETS:

Deadlift x5 (hip hinge / bilateral)

Power Push-Up x5 (upper body press / explosive)

TRX Knee Tuck / Body Saw x5+5 (core / unstable)

If you prefer to integrate the Full Combo idea with hangboard training, a simple idea is to structure it in simple bi-sets. Lots

of people have a hard time moving away from the idea of just standing below the hangboard for an entire session. A great set-up is like this:

3 ROUNDS OF EACH GROUP:

GROUP 1:

1-Arm Inverted Row x5+5 (upper body pull / unilateral)

Edge Hang x8 seconds

GROUP 2:

Rack Squat x5 (squat / bilateral)

Pocket Hang x8 seconds

GROUP 3:

Single Arm Kettlebell Swing x5+5 (hip hinge / unilateral + explosive)

Pinch Hang x8 seconds

GROUP 4:

Hardstyle Plank x10 seconds (core / isometric)

Edge Hang x8 seconds

GROUP 5:

Iso Overhead Press x5 seconds (upper body press / isometric)

Pocket Hang x8 seconds

There is no need to change the exercises as often as listed above. This example is just used to train a bit of breadth for an athlete who has mastered several variations of each exercise. If you haven't mastered an exercise in each category, the smart move is to avoid programming them.

4 Week Example of Variants:

UPPER BODY PRESS	WEEK 1 UNILATERAL	WEEK 2 EXPOSIVE	WEEK 3 BILATERAL	WEEK 4 UNSTABLE
Upper Body Pull	Explosive	Bilateral	Unstable	Isometric
Squat	Bilateral	Unilateral	Isometric	Unilateral
Hinge	Unstable	Isometric	Unilateral	Explosive
Core	Isometric	Unstable	Explosive	Bilateral

INTEGRATED BOULDER

The Integrated Boulder sessions are similar in structure to integrated strength, except instead of simply hanging from an edge for finger strength, you'll perform a couple of boulder problems. We also adjust the resistance exercises to more specific movements or loading. The idea here is to keep developing high levels of strength in the weight room, but to get back into the mode of movement.

We use strength-focused boulder problems in these sets. A strength-focused problem is one where difficult holds, hard body positions, or tension are the limiters. Avoid power-oriented, balance, or excessively long problems. Steeper walls with small holds rule here, such as the Tension Board or Kilter Board. Remember, the idea is to continue to develop finger strength.

For resistance efforts, you can opt to do the same exercises as in the General Strength phase, or move to more specific movements. This is an excellent time to train specialized resistance, much like we do in the Full Combination sessions. I like the following three forms of exercises:

1. **Unilateral Strength:** Exercises done with one limb or side at a time. These are typically more core-intensive than bilateral exercises and help develop balanced strength.

2. **Isometric Strength:** Holding static positions under load. Holding a difficult position is a great way to build strength quickly, but ideally you'd do just one or two isometric exercises per session. 5-10 seconds holds are a good place to start, and a drive to exert maximum force is key to progress.
3. **Concentric-Only Strength:** In climbing, we are often pulling down hard under load, and then reach up with an unloaded arm to grab the next hold. This is a different load than we apply when doing typical exercises. An example of training this in the gym would be to do a pull-up, but instead of lowering from the top, stepping off onto a box. This is less taxing than a full up-down cycle, but strength gains are similar. With a little thought and creativity, you should be able to come up with concentric-only exercises for most movements.

A TYPICAL INTEGRATED BOULDER GROUP LOOKS LIKE THIS:

2x Strength-Focused Boulder Problems
Unilateral, Isometric, or Concentric-Only Exercise
Mobility or Flexibility Drill

THE SESSION SET-UP WOULD LOOK LIKE THIS:

3 SETS OF EACH GROUP, DONE CIRCUIT-STYLE:

A1: 2x Strength-Focused Boulder Problems

A2: Deadlift with Drop, Mid-Thigh Pull, or Single Leg Deadlift, 2-5 reps

A3: Hip Mobility, 60 seconds

B1: 2x Strength-Focused Boulder Problems

B2: 1-Arm Push Up, Isometric Bench, or Single Arm Overhead Press, 4-6 reps

B3: Shoulder Mobility, 60 seconds

C1: 2x Strength-Focused Boulder Problems

C2: Pistol Squat, Step-Up, or Squat Hold 4-6 reps

C3: Hip Mobility, 60 seconds

All told, you'd complete 18 boulder problems throughout the session. The climbing and the switching of shoes, not to mention the moving around the gym takes some time. If your sessions are limited to an hour or so, you can shorten these workouts by dropping from 3 sets of each group to two.

COMBINED CIRCUITS

Combined Circuits are basically big Integrated Strength circuits. They are more specific to performance in that the volume of training taxes your capacity, and the variety of movement more closely simulates the metabolic demands of climbing. This is a good format for 2 or more people who want to train together using the same exercises. The big drawback is that you are probably going to be using several pieces of equipment at once and this can be impractical in a busy gym.

My recommendation is to plan on doing 1-2 circuits per session, and time them instead of doing them for a fixed number of rounds. Whether you make it through each exercise three or four or five times doesn't matter as much as doing them with high loads and executing them well. With more than one person training together, everyone can then go at their own pace.

A side goal of the Specific Strength phase is to continue to develop work capacity with a mind toward the phases coming later in the year. Paying attention to your total work per circuit will help you see that this capacity is increasing. In addition to tracking sets, reps, and weight, track how many times you make it through each circuit. Although you should not pursue increasing this number (that is something we'll do in the power endurance sessions), you will likely see it creep up slightly across the 4 weeks of the phase.

The circuits should be made up of 5-10 exercises, should include big muscle movements, mobility, and climbing-specific exercises, and you should set a work time of 15-20 minutes each. If you are doing a single large circuit, its duration can be 30-40 minutes. Simply move through the circuit of exercises at your own pace for the duration planned, resting as needed. Keep track of the rounds / exercises

1. Warm-Up. Kettlebell Complex (see Power Endurance section) 3 rounds.

2. Circuit 1 (15 minutes)

Upper Body Pull 4-5 reps

Squat 4-5 reps

Hangboard 5 seconds hang, rest 10 seconds,
5 seconds hang

Hip Mobility

Levers or Straight Leg Raises 4-5 reps

Campus Ladder

3. Circuit 2 (15 minutes)

Upper Body Press 4-5 reps

Hip Hinge 4-5 reps

Hangboard 5 seconds hang, rest 10 seconds,
5 seconds hang

Shoulder Mobility

Ab Wheel or Dragon Flag 4-5 reps

Campus Ladder

Circuits can trick us into seeking fatigue. If you are sweating heavily, breathing hard, or are getting pumped, you're out of the strength zone. It's likely that your loads are too light, and your rests are too short. At the end of an ideally-loaded circuit, you might feel "powered down" but should not feel like puking. If you



PHOTO: NATE LILES

are super pumped or super sweaty at the end of one, adjust the loads up next time, and force more rest. We'll do lower-intensity and higher-fatigue circuits when the time comes...in the Power Endurance phase.

GO A HUNDRED FINGER STRENGTH

The major complaint we get about most of our hangboard programs is that they feel too easy. Well, we'd way rather get strong without suffering than suffer without getting strong. That's where this plan really shines - it's simple, progressive, and gets you strong over the long haul.

Go A Hundred is simple. You hang for 100 seconds on an edge, then progress slowly to smaller and smaller holds. This plan is dependent on using a board with multiple hold sizes, such as the Tension Climbing Grindstone Board or the Transgression/Progression Board. It works reasonably well with a campus or system board with multiple edge sizes, but we've found you need about 5 progressively smaller edges to make it work.

Hang sets should be between 5 and 20 seconds. At the beginning of each cycle, you'll be using bigger holds and will trend toward the 20 second end of things. As the cycle progresses, you'll be on the smaller holds, and will end up having to do the sets for shorter durations. If you get to where you can't do 5-second hangs, you're past the point of useful work and increasing your chance of injury...especially at a volume of 100 seconds per session. At this point, you should opt for a bigger series of holds.

Yes, a 20 second load addresses muscular endurance more than strength when it comes to adaptation. I like 20 second loads, though, because we are in a safe zone for the connective tissue in the fingers and because this represents a more real-world load - when we are climbing many of us tend to hang on for longer than 5 seconds per hold. If you think this seems crazy, don't email about it...just stick with doing repeaters.

The sessions can be done by integrating them with other strength work, or simply as straight sets. Rest as needed. Ahh, yes... “rest as needed.” Many people worry “how much time, exactly, should I rest between sets?” It doesn’t matter much, but if you’re getting pumped as you go through the session, it’s not enough rest and if you eat a meal between sets, it might be too much. If you’re honestly not sure, go for 3-5 minutes.

The metric should not be how tired the training makes you, but rather how strong.

One recommendation I’ll make is to do a quick set of finger extensions (I like the Expand Your Hands Band) for 10 reps per hand. There is good science behind activating antagonists to increase strength in the agonists, but honestly, I just think it makes the fingers feel better!

An example session, integrated with weight training in a Combined Circuit:

5 ROUNDS OF:

Deadlift x 2

1-Arm 1-Leg Push Up x 2 each side

Pistol Squat x 2 each side

DB Row x 2 each side

Levers x 2

Edge Hang Half Crimp x 20 seconds (or less!)

Extensions x 10 each side

If one didn’t get all 100 seconds on the hang, he’d add a few sets at the end. No big deal. Yes, we are only training one hold position. Although I don’t feel it’s necessary for most climbers, you can add supplemental positions (at significantly reduced volumes). We suggest doing 20-30 seconds of total loading on supplemental positions, in 5-10 second sets. A nice way of setting this up would be to cycle in maybe two other positions in an alternating fashion, like this:

Half Crimp Edge 20 seconds
Hang Pocket 10 sec

Half Crimp Edge 20 seconds
Hang Pinch 10 sec

Half Crimp Edge 20 seconds
Hang Pocket 10 sec

Half Crimp Edge 20 seconds
Hang Pinch 10 sec

Half Crimp Edge 20 seconds
Hang Pocket 10 sec

As always, we'll remind you that the specifics don't matter as much as *regular* loading over several weeks and *progression* through a long cycle. Dialing in some killer session that you only do once or twice in a season is a waste of time. Although it might make you tougher and boost your ego, it won't make you stronger. Also... strength ramped up quickly tends to show a steep decline on the back side, too. The longer and more deliberate your strength, the longer-lasting it will be.

CYCLE 1

I have abbreviated this cycle from the one I originally wrote for the Climb Strong site, so it is 4 weeks in duration rather than 5. If you have not just finished a General Strength phase, do Cycle 1 in the Specific Phase. This is the one you'll feel like you don't need. You're experienced, you just sent 13c, and you've got a "good base" - whatever that means. If you want to get stronger and you want to avoid injury, please don't disregard this cycle. Each time you do this plan, you must do cycle 1 - it's the part that keeps you from getting hurt.

Start with the largest edge on the board. On the Grindstone, this will be 35mm. This workout should just involve doing 5 sets of 20 second hangs. Do this for one session. Don't add load, just do it at bodyweight, keeping it simple and easy.

Session 2 you will move to the next smaller edge, which is the 30mm on the Grindstone. This is a bit of a leap, but most climbers will still be able to stick with the 20 second sets. Session 3 will see you on the 25mm edge, which you'll do for one session.

Sessions 4 and 5 are done on the 20mm edge, then sessions 6 and 7 on the 15mm. On the final session, you can opt to go to a smaller edge (10mm), or stick with the 15mm. Many climbers will be able to perform 20 seconds per hang straight through all 8 sessions. Good job.

If you need to break these down to 15 second sets, 10 or even 5, that's fine. It just shows us that you'll move through the whole plan more quickly. Here is a review of Cycle 1 in table form with some example hold sizes:

SESSION #	EDGE SIZE (HALF CRIMP)	(GRINDSTONE)	(PROGRESSION)
1	Largest	35mm	Top Edge
2	Second-Largest	30mm	24mm
3	Mediumest	25mm	20mm
4 and 5	Second-Smallest	20mm	16mm
6 and 7	Smallest	15mm	12mm
8	Super Smallest	10mm	10mm

CYCLE 2

You may have come out of a really good General Strength phase, and are now lining up for 4 weeks of specific strength building. You may have even done a Go A Hundred Cycle 1 recently. If you made it through the whole progression on that cycle at 20 seconds per hang, add 20% to your bodyweight for this series. If you dropped down to shorter-duration sets, but were able to stay above the 5 second mark, add 10%. If you were working at 5 second sets by the end, stick with bodyweight.

I'll underscore that I don't feel micro-adjusting load matters. The effective zone for strength gain in isometrics is between 70% and 100% of maximum voluntary isometric contraction, or "max." Therefore, if you weigh 150 pounds, and your 1-second max on a given edge is 200 pounds (bodyweight + 50 pounds), The effective training load for strength gains is anywhere between 140 and 200 pounds. This means you can add almost any load (or even reduce load by 10 pounds, in this case) and still see gains. Yes, there are optimal zones, but the take-home is that you don't need to get too fancy with your exact loading. Micro-adjusting serves your organized mind more than your muscles.

You will use the exact-same load for the whole cycle, so be conservative.

PERFORMANCE LAST PHASE	LOAD FOR CYCLE 2
All sessions at 20 seconds	Bodyweight + 20%
All sessions at more than 5 seconds	Bodyweight + 10%
Some sessions at 5 seconds	Bodyweight

We are tempted to try to up the ante every time we do well in strength training. In this plan, though, the improvements come

by progressing to smaller holds and bringing the loads up slowly. Think long-term instead of next week. Think Roth IRA instead of Vegas Strip.

Cycle 2 will feature more rest between sessions. I recommend at least one day between. Again, you can put these in with other strength work, with a mobility session, or even with some bouldering or climbing. Here is a session of Integrated Strength one of my athletes did recently in cycle 2:

KB Swing x 10, Overhead Press x 5, Straight Leg Raise x 5,
Hang 15mm (+20%) x 15 sec
KB Swing x 10, Overhead Press x 5, Straight Leg Raise x 5,
Hang 15mm (+20%) x 13 sec
KB Swing x 10, Overhead Press x 5, Straight Leg Raise x 5,
Hang 15mm (+20%) x 15 sec

Rack Step-Up x 5+5, Pull-Up x 5, Ab Wheel x5, Hang 15mm
(+20%) x 12 sec
Rack Step-Up x 5+5, Pull-Up x 5, Ab Wheel x5, Hang 15mm
(+20%) x 13 sec
Rack Step-Up x 5+5, Pull-Up x 5, Ab Wheel x5, Hang 15mm
(+20%) x 12 sec

TGU x 1+1, Plank Rope Pull x 10m, Frog x 45 sec, Hang
15mm (+20%) x 12 sec
TGU x 1+1, Plank Rope Pull x 10m, Frog x 45 sec, Hang
15mm (+20%) x 8 sec
TGU x 1+1, Plank Rope Pull x 10m, Frog x 45 sec (no hang,
100 sec already done)

You will follow the exact-same progression as Cycle 1. Typically, the first few sessions will go easily, with most hangs at close to 20 seconds. Later in the phase, the sessions can be real grinders. In

the latter phases, you may end up with sets of 5-10 seconds, split by lots of rest, for 10-20 sets...a serious day's work.

SESSION #	EDGE SIZE (HALF CRIMP)	(GRINDSTONE)	(PROGRESSION)
1	Largest	35mm	Top Edge
2	Second-Largest	30mm	24mm
3	Mediumest	25mm	20mm
4 and 5	Second-Smallest	20mm	16mm
6 and 7	Smallest	15mm	12mm
8	Super Smallest	10mm	10mm

MAX HANGS + MAX PULLS

This style of session is what most of us probably did when we first thought of training for harder climbing. We did pull-ups and we hung by our fingers. Although lacking in a lot of depth, this line of thinking does lead to really strong fingers and pulling muscles. And despite the fact that you might be neglecting some other movements, remember that this is just 8 workouts - you probably won't lose too much in the areas you're not focused on.

The person who is most attracted to this session probably needs it the least. We do Max Hangs and Max Pulls to really develop upper body strength. If you're a guy that likes to whip out a couple of one arms anytime you're near a pull-up bar and have an audience, this might not be the plan that moves you the furthest forward. If you have problems with lock-offs, specific hold types, or steep climbing, this may be a perfect solution. However,

you must know your way around the weight room and understand training with percentages. If you are not there, this is not the workout for you.

In this session, you'll combine pull-ups, row variations, and hangs on the hangboard, all at high loads. Where most of the strength training in this book is done at the 70-90% intensity level, these sessions must be done at 85%+ of max to be most effective. This means hangs of 5 seconds or less, and pulls at 3-rep max or heavier.

The warm-up should include movement preparation for 5-10 minutes followed by another 5-10 minutes of work with some slowly increasing durations on the hangboard and a few pull-ups. Then, you can ramp up the pulls with dumbbell rows, doing 5+5 at 70%, 4+4 at 80%, and then 3+3 at 85 or 90%.

GROUP A (5 ROUNDS):

Pull-up x 3 (@85%)

Hang, medium edge x 5 seconds (@85%+)

DB Row x 3+3 (@85%)

Hang, medium edge x 5 seconds (@85%+)

- Rest 1-3 minutes between each exercise. A good rule of thumb is that if you can sustain high loads at less rest than this, the loads really aren't that high.

GROUP B (3 ROUNDS):

Pull-Up Iso Hold (@90 degree elbow) x 5+5 seconds

Hang, small edge x 5 seconds (@85%+)

Inverted Row x 5+5

Hang, small edge x 5 seconds (@85%+)

This is a taxing session. Rest two days between these sessions, no matter what. A typical week will be Sunday: Hangs and Pulls, Wednesday: Hangs and Pulls, Friday: Maintenance. If the elbows or shoulders start to hurt, it's an indication your training before



PHOTO: JOSH LARSON

this phase was not sufficient to support this loading. Back off and come back to it next time around.

POWER TRAINING SESSIONS

GENERAL POWER SESSIONS

3X2 POWER

The 3x2 Power workout is a quick and very effective way to ramp up your power. Each group consists of one lower-body power exercise, one upper body power exercise, and one mobility focus. Power exercises are typically high-skill movements, so if you're not sure how to perform them, don't do them until you have found a good coach. Rest 1-2 minutes between rounds, even if you feel like you don't need it.

It is possible to do this work after a bouldering session or even on the following day. If you are moving really high loads in the weight room, it's not a good idea to climb afterward due to nervous system fatigue.

GROUP A (3 ROUNDS):

Box Jump x5-8 reps

Power Push-Up x5-8 reps

Hip Mobility (Frog or Firelog) x60-90 seconds

GROUP B (3 ROUNDS):

Barbell Jump Squat x 3-4 reps

Snatch (DB or KB) x5-8 reps each side

Shoulder Mobility (Arm Bar or Superband Shoulder Stretches) x 60-90 seconds

INTEGRATED POWER

Integrated Power is similar to integrated strength except that you use power drills instead. This is my favorite general power session. When training power, I find the hardest thing about the sessions is resting enough, so the integrated format works really well to keep me on track. The same general set-up is seen here as in strength:

3 SETS OF EACH GROUP, DONE CIRCUIT-STYLE:

A1: Hangboard position 1

A2: Box Jump or Barbell Jump Squat, 2-4 reps

A3: Hip Mobility, 60 seconds

B1: Hangboard position 2

B2: Push-Press or Power Push-Up, 5-8 reps

B3: Shoulder Mobility, 60 seconds

C1: Hangboard position 3

C2: DB or KB Snatch, 5-8 reps each side

C3: Hip Mobility, 60 seconds

I stick with hangboard in the general phase rather than transitioning to movement-based power such as campusing. This is to help reinforce the strength gains from the last phase, as well as to keep from going into Specific Power too quickly. Remember, you will still be bouldering and won't be lacking in opportunities to be powerful there.

THE COMPLEX METHOD

The Complex Method takes advantage of an interesting finding from sports science (a good place to start learning about this is *Designing Resistance Training Programs* by Fleck and Kraemer) in which some athletes have seen slightly better results in their power development if the power exercises were preceded by a strength

set. This is thought to occur because of a mechanism called *post-activation potentiation*. For example, instead of simply doing a set of box jumps, you'd do a set of heavy deadlifts, then walk over to the box and do your jumps. The combo would then be repeated a few times. Although this is an advanced training method and might or might not produce superior results for you, the combo of strength exercises and power exercises provides good benefits. It allows for continuity between the movements developed in the strength sessions last phase and helps reinforce the strength levels gained. It also helps control the amount of volume you put into any one exercise, since for every power drill you'll be doing a strength exercise, too.

A standard workout format is as follows:

GROUP A (4 ROUNDS):

Deadlift x2-3

Box Jump x4-6

Hip Mobility x60-90 seconds

GROUP B (4 ROUNDS):

Bench Press x2-3

Power Push-Up x4-6

Shoulder Mobility x60-90 seconds

This workout will feel short and will feel easy until you start increasing the loads a bit. Trust the session, though. Your numbers will slowly climb and the stated goal of the phase, to develop general explosiveness, will come.

SPECIFIC POWER SESSIONS

INTEGRATED BOULDER

Integrated Boulder training sessions work on the same idea behind the Integrated Strength format. By combining specific and

powerful climbing movement with non-specific highly-focused explosive efforts, we see two major benefits:

1. The athlete gets more out of the general explosiveness training because climbing between the hard sets assures adequate rest periods, and,
2. Doing high-intensity explosive efforts in the weight room assures progress in power development, where just bouldering alone might not.

The session formats can be built several ways. Depending on your training facility set-up, your ability to move between a bouldering wall and a training area might vary. Here are a couple of example sessions:

INTEGRATED BOULDER A

1. 5-10 minutes of a standard Bouldering Warm-Up
2. 4 rounds of:
 - 3x Power-Oriented Boulders, Onsight Level or so, resting as needed between
 - 5x Ball Slam, rest 30 sec
 - 5x Box Jump, rest 30 sec
 - 3x Power-Pull-Up, rest 4 minutes.
3. 4 rounds of:
 - 2x Power-Oriented Boulders, Onsight Level or so, resting as needed between
 - 5x Power Push-Up, rest 30 sec
 - 5+5x Kettlebell Snatch, rest 30 sec
 - 2x Power-Pull-Up, rest 4 minutes.

(Sets of boulders and pull-ups reduced in the second group to ensure adequate rest and quality of movement)

INTEGRATED BOULDER B

1. 5-10 minutes of a standard Bouldering Warm-Up
2. 45 minutes of Power-Oriented Boulders, approximately one boulder every 4 minutes, resting more if fatigue is an issue or if power declines. Approximately onsight level.
3. Rest no more than 30 minutes, transition to weight room.
4. 5 rounds, start each exercise at the top of the minute (each round will take 6 minutes):
 - 5x Power Push-Up
 - 5x Box Jump
 - 2x Power-Pull-Up
 - 5+5x Kettlebell Snatch
 - 5x Ball Slam
 - Rest full minute

(Some athletes complain that this circuit feels too easy and might try to “intensify” it by adding more reps or reducing rest. Both are counter to the goal of building power. If it feels too easy, add load or increase speed.)

BOULDER / CAMPUS

The Boulder / Campus sessions are a style of training most of us are pretty familiar with, but I want to make the goal of these sessions crystal-clear. You are interested only in power output, which leads to power development. Too often, we work our way through an extended boulder session and think that we are still powerful simply because we continue to improve on the moves. More often than not, this is a function of improved efficiency and learning, and we can continue to get better despite loss of power.

In this session, fatigue has no place. Remember, fatigue creates endurance and is the enemy of power.

Once again, your facility set-up is going to be the main variable in this training. Most climbers do best when they can switch between the bouldering wall and the campus board more frequently in the session. If you can't do it, simply stick to the rule of "rest a lot, stop when power drops," and you can train the two separately.

Our goal with the bouldering is to keep chasing explosiveness and contact strength. Aim to stay on steep terrain that requires big moves and deadpoints. Avoid problems that require you to move slowly or feature small holds as the main limiters.

1. 5-10 minutes of a standard Bouldering Warm-Up

2. 6-10 rounds of:

- 2x Power-Oriented Boulders, Onsight Level or OS+1, resting as needed between
- 1+1x Campus Ladder (approximately 8 seconds in duration. The 1+1 designation indicates you'll do one ladder leading with the left hand to start, followed by a second one mirroring the movement with the right hand leading. Rest 45-90 seconds between these ladders. Think explosiveness here. Big moves on big rungs are more useful in this session than small rungs.

If you start to get fatigued, slow your pace. Each problem and each trip up the Campus Board should feel fresh.

ENDURANCE TRAINING SESSIONS

GENERAL ENDURANCE SESSIONS

CARDIAC OUTPUT

We all know about “cardio” and its overall benefits to health. We also know that our bodies show some of the same responses to hard climbing as they show to difficult cardiovascular training, such as sweating, fatigue, increased heart rate, and labored breathing. The idea of doing more of *this* to get better at *that* is not a tough connection for most of us to consider, yet we have to be cautious. Simply adding in several hours of running or cycling per week will not magically increase your ability to avoid getting pumped.

Although I argue against specifically using cyclic endurance exercises (running, cycling) to build climbing endurance, Cardiac Output training has its place in your conditioning program. The cardiac output modes can be just about anything that increases the heart rate and is sustainable for 20-90 minutes, but a few guidelines should be followed. By holding the intensity fairly low, your adaptations differ from harder efforts. Long, slow training increases the stroke volume of the heart, which results in *eccentric cardiac hypertrophy*. This, in turn, improves cardiac efficiency, decreases resting heart rate, and decreases working heart rates at any given level of work. Higher intensity exercise (tempo-paced efforts or exercising close to anaerobic threshold) result in concentric hypertrophy - and instead of increasing stroke volume will increase the heart's ability to exert more pressure...essentially by increasing heart wall thickness and size. We don't need this high-level of cardiac development for climbing.

If we do interval-style efforts, we train the heart to contract quickly, often before the chambers fill completely with blood. This doesn't allow for the eccentric overload we are looking for. What we need is lots of slow, steady activity, preferably using the whole body. Don't address this as training for _____ endurance activity. You are only using this activity to build base conditioning for climbing.

For Cardiac Output sessions, follow these guidelines:

- Do them 1-3x per week. If you are challenged by these efforts, more frequent and shorter sessions are the key. These can be done on the same day as other training if desired.
- Exercise for 20-90 minutes in as close to a non-stop mode as possible.
- Cross-country skiing, swimming, rowing, or machine training that requires both upper and lower body involvement (SkiErg, air bike, rowing machine) are the best. Hiking or easy jogging are OK.
- Maintain conversational intensity (being able to speak in full sentences), or a heart rate of 120-150 beats per minute.
- Increase the training effect by adding more sessions or longer durations - not by increasing speed.

A final note: don't try to add more of this kind of training in order to lose weight or to give yourself more leeway on your diet. If you need to lose fat, do so through diet interventions and habit changes.

ALACTIC CAPACITY CIRCUITS

We all understand that 4x4 intervals wreck us and we get super sweaty and pumped doing them. My question is this: did my endurance improve? The answer is a “probably yes,” but 4x4s only attack one half of the endurance equation. Instead of getting better at tolerating pain, at climbing while wickedly pumped, we should focus the majority of our training on teaching our bodies not to get pumped in the first place. By focusing our efforts on improving the capacity of our alactic energy system (the one responsible for strength and power efforts), we can effectively set a great base level upon which to train power endurance sessions later in the season.

Alactic Capacity circuits feature high-intensity efforts performed for 5-10 seconds, and followed by passive rest. Sport science taught us that these efforts done on a 1:5 or longer work:rest ratio were best, so for many cycles we trained by doing a 5-10 second effort each minute. What we slowly realized was that alternating the focus of the efforts from finger-specific to total-body movements allowed us to cheat the work:rest ratio a bit, and thus the intervals on a 30 second clock were born.

In the General Endurance phase, we lean more toward total body efforts. During the Specific Endurance phase, we'll continue the Alactic Capacity circuits, but aim toward more specific movements. Each set of intervals features 5 exercises done in sequence. You start each exercise at the beginning of a :30 repeating clock, then rest the remainder of the time. In general this gives you about 20 seconds to move to the next exercise. After the 5th exercise, you'll rest an additional 30 seconds, thus making each round 3 minutes in length. The general template is as follows:

0:00 >10 seconds Upper Body Explosive

0:30 >10 seconds Lower Body / Total Body Explosive

1:00 10 second Edge Hang

1:30 >10 seconds Upper Body / Total Body Explosive
2:00 >10 seconds Upper Body Strength
2:30 rest full 30 seconds

There is not a lot of time for moving around the gym here. I suggest you grab a couple of tools and set yourself up by the Campus Board. My go-to session is the following:

0:00 Campus Ladder 1-3-5-7-9 (on biggest rungs)
0:30 6x Kettlebell Swings (~½ bodyweight bell)
1:00 10 second Edge Hang (bodyweight, 12mm)
1:30 5x Ball Slam (light ball, go for speed)
2:00 2x Power Pull-Up
2:30 rest full 30 seconds

After one round you should feel pretty good. Used some power, not sucking wind. The progression of loading in the General Endurance phase is to add more total work in the form of more rounds. Increasing intensity, decreasing rest, or adding duration to the sets will start to force you into the glycolytic zone and cause a power-endurance peak, which will not serve you at this point in the year.

The sessions are built on doing several rounds of 3 minutes, as described above. A series of rounds will be done back-to-back (usually 3-6 rounds per series), with a long rest between series. It is possible to change a few of the exercises between series, but don't get carried away with variety.

You must continually assess whether your strength and power are staying up. If you start to see a notable decline in output or in exercise form, it's indicative you've reached the end of the effective session length. Going deeper into the training will be counter-productive. In the progressions below, understand that almost all of us will hit a level where we should not add more work, and should stick with the same session for a week or two. This



PHOTO: JULIE ELLISON

session is appropriate once per week. If you decide to do this style of session more frequently, only advance the session weekly, not each session.

Being a grown-up about it rather than charging forward into tiredness will produce better long-term results.

Progression is as follows:

Session 1: 2 series of 4 rounds with 5 minutes between.
(29 minutes total)

Session 2: 2 series of 5 rounds with 5 minutes between.
(35 minutes total)

Session 3: 2 series of 6 rounds with 5 minutes between.
(41 minutes total)

Session 4: 3 series of 5 rounds with 5 minutes between.
(55 minutes total)

If your form breaks down or you are unable to complete an exercise, don't follow the progression. Back off on the exercise, reduce the number of reps per set, or reduce the load. Your feeling at the end of any of these sessions should be tired and powered down, not nauseous and pumped.

I recommend these sessions across both the General and Specific Endurance blocks. If you choose to include it for a full 8 weeks of training, you should follow the progression above, then repeat it exactly in the second 4 weeks but with increased loads. The whole idea here is not to increase your ability to handle highly glycolytic (pumpy) sessions, but rather to complete a lot of hard and powerful work without going into that zone.

EXTENSIVE ENDURANCE INTERVAL COMBINATIONS

Extensive Endurance is simply the endurance zone where you can climb continuously for a full pitch worth of climbing without going into the anaerobic zone. This is the zone that climbers were trying

to target in the early days of training with SACC training, and is probably slightly more fatiguing than ARC training is supposed to be, as ARC stands for Aerobic Restoration and Capillarity. ARC was modeled after cyclists' easy zone 1 rides where they flushed out the system with low intensity activity. I don't believe that any climbing can be easy enough to act as a restorative for any but the most elite performers. Extensive Endurance (EE) climbing would be better equated with "zone 2" training: hovering around *aerobic* threshold (as opposed to anaerobic), or more simply, conversational intensity.

The big problem with any endurance climbing at super-low intensities is that the skin and joints become a legitimate issue. Your ability to keep going starts to really decline when your tips say "no" to each hold you grab. To combat this, the EE Combinations bring in general total body endurance exercises, like those recommended in the Cardiac Output sessions, to help you develop capacity and keep the heart rate up between bouts of low-intensity climbing.

The climbing portion of these sessions can be done in a number of ways. The big key is to try to actually be moving for as much of the interval as possible. This means avoiding rest positions and shakeouts, and it also means trying to limit lowering and transition times if climbing on a rope. Ideally, a climber doing routes would either downclimb or lower quickly and immediately start climbing again to try to keep the intensity of the work steady. Roped climbing is not ideal because the gear management and transitions are time-consuming.

I also caution against mindless open traversing on a bouldering wall. We tend to fall into bad patterns here. Although there are sessions where simply burning out the forearms can be productive, it should not form a large bulk of your training, and should not be done in a skilled (real climbing) environment. If you practice dumbbed-down movement, you'll start using it when you don't want to.

As far as the nonspecific Cardiac Output modes go, almost anything can work, yet I recommend doing an activity that involves the whole body rather than just the legs. Machines such as the Ski Erg, rowing machine, or dual-action bike are good choices, as are low-intensity weight circuits. The key will be to hold the difficulty of these modes to a very low level, at conversational pace - you should be able to speak in full sentences without gasping while doing these intervals. A second rule of thumb would be this: if you need a rest after the nonspecific exercise before you start climbing again, you went too hard.

1. 5-10 minutes, bouldering volume ladder. Do an easy boulder problem, then rest 30-45 seconds. Next, do 2 easy problems with no rest between, then rest 30-45 seconds. Follow this with 3 problems back-to-back. If needed, repeat this sequence a second time. Rest 5 minutes before starting training.
2. Intervals. Have your equipment set-up and “reserved” so you can go right there after climbing. Find a good transitioning area in the gym and set out your climbing gear and your training shoes there. I recommend 2 different progressions below. Even if you are a high-level climber, but have not done this style of training before, you are advised to start with the low-volume progression.

Please note that these are suggested progressions. You have to be a grown-up when you plan your own training. If the effort in a certain workout really hammers you, chances are you shouldn't progress that session...it's OK to simply do the same volume for a few weeks. Likewise, if you can maintain high skill movement and are not fazed by the training at all, adding some volume might be appropriate.

I'll reiterate that these sessions should not destroy you. They should not get you pumped or make you sore or require rest days

after. Hold back on the throttle until the power endurance workouts later in the year.

Low-Volume Progression

SESSION #	NUMBER OF ROUNDS	CLIMBING TIME (MIN)	TRANSITION TIME TO CO TRAINING	CO TRAINING TIME (MIN)	REST BETWEEN ROUNDS	TOTAL TIME (MIN)
1	3	5	1	5	5	43
2	3	6	1	5	5	46
3	3	7	1	5	5	49
4	2	5	1	4	5	55
5	4	6	1	4	5	59
6	4	7	1	4	5	63

(MIN) = in minutes

High-Volume Progression

SESSION #	NUMBER OF ROUNDS	CLIMBING TIME (MIN)	TRANSITION TIME TO CO TRAINING	CO TRAINING TIME (MIN)	REST BETWEEN ROUNDS	TOTAL TIME (MIN)
1	4	8	1	6	4	72
2	2	10	1	5	4	76
3	4	12	1	5	4	84
4	5	8	1	4	4	80
5	5	10	1	4	4	90
6	5	12	1	4	4	100

(MIN) = in minutes

SPARTACUS CIRCUITS

Spartacus circuits are workouts we’ve been doing in our gym for the past 10 years or so. These were born from a total-body weight circuit we learned from strength coach Alwyn Cosgrove, and were